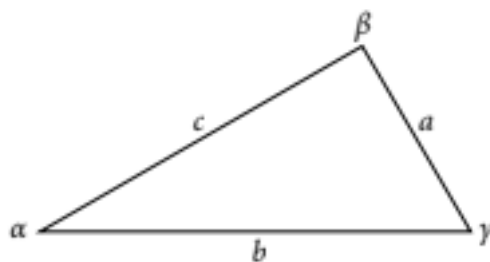


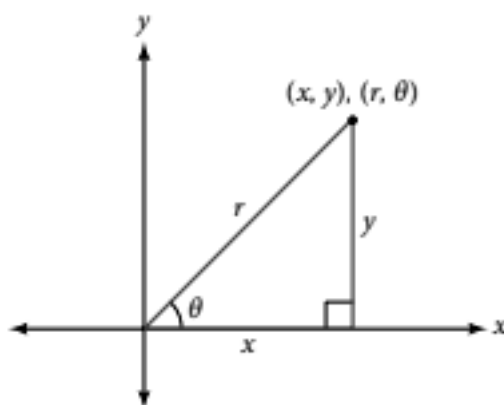
# EXAM III EQUATION SHEET

|                                   |                                                                                         |                                                          |                                                          |
|-----------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|
| Even-Odd Identities               | $\tan(-\theta) = -\tan(\theta)$                                                         | $\csc(-\theta) = -\csc(\theta)$                          | $\sin(-\theta) = -\sin(\theta)$                          |
|                                   | $\cot(-\theta) = \cot(\theta)$                                                          | $\sec(-\theta) = \sec(\theta)$                           | $\cos(-\theta) = \cos(\theta)$                           |
| Reciprocal Identities             | $\sin(\theta) = \frac{1}{\csc(\theta)}$                                                 | $\csc(\theta) = \frac{1}{\sin(\theta)}$                  |                                                          |
|                                   | $\cos(\theta) = \frac{1}{\sec(\theta)}$                                                 | $\sec(\theta) = \frac{1}{\cos(\theta)}$                  |                                                          |
|                                   | $\tan(\theta) = \frac{1}{\cot(\theta)}$                                                 | $\cot(\theta) = \frac{1}{\tan(\theta)}$                  |                                                          |
| Quotient Identities               | $\sin(\theta) = \frac{1}{\csc(\theta)}$                                                 | $\csc(\theta) = \frac{1}{\sin(\theta)}$                  |                                                          |
|                                   | $\cos(\theta) = \frac{1}{\sec(\theta)}$                                                 | $\sec(\theta) = \frac{1}{\cos(\theta)}$                  |                                                          |
|                                   | $\tan(\theta) = \frac{1}{\cot(\theta)}$                                                 | $\cot(\theta) = \frac{1}{\tan(\theta)}$                  |                                                          |
| Pythagorean Identities            | $\cos^2(\theta) + \sin^2(\theta) = 1$                                                   | $1 + \cot^2(\theta) = \csc^2(\theta)$                    | $1 + \tan^2(\theta) = \sec^2(\theta)$                    |
| Sum/Difference Formulas (Cosine)  | $\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$              |                                                          |                                                          |
|                                   | $\cos(\alpha - \beta) = \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta)$              |                                                          |                                                          |
| Sum/Difference Formulas (Sine)    | $\sin(\alpha + \beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta)$              |                                                          |                                                          |
|                                   | $\sin(\alpha - \beta) = \sin(\alpha)\cos(\beta) - \cos(\alpha)\sin(\beta)$              |                                                          |                                                          |
| Sum/Difference Formulas (Tangent) | $\tan(\alpha + \beta) = \frac{\tan(\alpha) + \tan(\beta)}{1 - \tan(\alpha)\tan(\beta)}$ |                                                          |                                                          |
|                                   | $\tan(\alpha - \beta) = \frac{\tan(\alpha) - \tan(\beta)}{1 + \tan(\alpha)\tan(\beta)}$ |                                                          |                                                          |
| Cofunction Identities             | $\sin(\theta) = \cos\left(\frac{\pi}{2} - \theta\right)$                                | $\cos(\theta) = \sin\left(\frac{\pi}{2} - \theta\right)$ | $\tan(\theta) = \cot\left(\frac{\pi}{2} - \theta\right)$ |
|                                   | $\sec(\theta) = \csc\left(\frac{\pi}{2} - \theta\right)$                                | $\csc(\theta) = \sec\left(\frac{\pi}{2} - \theta\right)$ | $\cot(\theta) = \tan\left(\frac{\pi}{2} - \theta\right)$ |

|                         |                                                                                                                                                                            |                                                |                                                                                  |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------------------------------------------------------------|
| Double Angle Formulas   | $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$                                                                                                                              |                                                | $\tan(2\theta) = \frac{2 \tan(\theta)}{1 - \tan^2(\theta)}$                      |
|                         | $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$                                                                                                                          |                                                | $\cos(2\theta) = 1 - 2 \sin^2(\theta)$<br>$\cos(2\theta) = 2 \cos^2(\theta) - 1$ |
| Reduction Formulas      | $\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$                                                                                                                             | $\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$ | $\tan^2(\theta) = \frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}$                   |
| Half Angle Formulas     | $\sin\left(\frac{\alpha}{2}\right) = \pm \sqrt{\frac{1 - \cos(\alpha)}{2}}$                                                                                                |                                                |                                                                                  |
|                         | $\cos\left(\frac{\alpha}{2}\right) = \pm \sqrt{\frac{1 + \cos(\alpha)}{2}}$                                                                                                |                                                |                                                                                  |
|                         | $\tan\left(\frac{\alpha}{2}\right) = \pm \sqrt{\frac{1 - \cos(\alpha)}{1 + \cos(\alpha)}} = \frac{\sin(\alpha)}{1 + \cos(\alpha)} = \frac{1 - \cos(\alpha)}{\sin(\alpha)}$ |                                                |                                                                                  |
| Product to Sum Formulas | $\cos(\alpha) \cos(\beta) = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$                                                                                     |                                                |                                                                                  |
|                         | $\sin(\alpha) \cos(\beta) = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$                                                                                     |                                                |                                                                                  |
|                         | $\sin(\alpha) \sin(\beta) = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$                                                                                     |                                                |                                                                                  |
|                         | $\cos(\alpha) \sin(\beta) = \frac{1}{2} [\sin(\alpha + \beta) - \sin(\alpha - \beta)]$                                                                                     |                                                |                                                                                  |
| Sum to Product Formulas | $\sin(\alpha) + \sin(\beta) = 2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)$                                                       |                                                |                                                                                  |
|                         | $\sin(\alpha) - \sin(\beta) = 2 \sin\left(\frac{\alpha - \beta}{2}\right) \cos\left(\frac{\alpha + \beta}{2}\right)$                                                       |                                                |                                                                                  |
|                         | $\cos(\alpha) - \cos(\beta) = -2 \sin\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$                                                      |                                                |                                                                                  |
|                         | $\cos(\alpha) + \cos(\beta) = 2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)$                                                       |                                                |                                                                                  |



|                 |                                                                           |                                             |                                              |
|-----------------|---------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------|
| Law of Sines    | $\frac{\sin(\alpha)}{a} = \frac{\sin(\beta)}{b} = \frac{\sin(\gamma)}{c}$ |                                             |                                              |
| Law of Cosines  | $\cos(\alpha) = \frac{b^2 + c^2 - a^2}{2bc}$                              | $\cos(\beta) = \frac{a^2 + c^2 - b^2}{2ac}$ | $\cos(\gamma) = \frac{a^2 + b^2 - c^2}{2ab}$ |
| Area            | $A = \frac{1}{2}bc \sin(\alpha)$                                          | $A = \frac{1}{2}ac \sin(\beta)$             | $A = \frac{1}{2}ab \sin(\gamma)$             |
| Heron's Formula | Let $s = \frac{a+b+c}{2}$ . Then, $A = \sqrt{s(s-a)(s-b)(s-c)}$           |                                             |                                              |



|                                                    |                                                                                                                                                                                                                                                                                                            |                              |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Polar/Rectangular Conversion                       | $\cos(\theta) = \frac{x}{r}$                                                                                                                                                                                                                                                                               | $\sin(\theta) = \frac{y}{r}$ |
|                                                    | $x = r \cos(\theta)$                                                                                                                                                                                                                                                                                       | $y = r \sin(\theta)$         |
|                                                    | $x^2 + y^2 = r^2$                                                                                                                                                                                                                                                                                          | $\tan(\theta) = \frac{y}{x}$ |
| Polar Form of Complex Number                       | $z = r \operatorname{cis}(\theta) = r [\cos(\theta) + i \sin(\theta)]$                                                                                                                                                                                                                                     |                              |
| Products/Quotients of Complex Numbers (Polar Form) | <p>Suppose <math>z_1 = r_1 \operatorname{cis}(\theta_1)</math> and <math>z_2 = r_2 \operatorname{cis}(\theta_2)</math>.</p> <p>Then, <math>z_1 z_2 = r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2)</math> and <math>\frac{z_1}{z_2} = \frac{r_1}{r_2} \operatorname{cis}(\theta_1 - \theta_2)</math></p> |                              |
| De Moivre's Theorem                                | Suppose $z = r \operatorname{cis}(\theta)$ . Then, $z^n = r^n \operatorname{cis}(n\theta)$ .                                                                                                                                                                                                               |                              |